KitBot build & wiring

Build with the video. Follow the picture wiring pages. Test before securing anything with glue.

Gather your parts

Controller. Wukong board with built-in battery, plus your BBC micro:bit.

Drive. Two Geekservo 360-degree motors, two motor cables, wheels and chassis parts.

Sensors. HC-SR04 ultrasonic sensor and TCS34725 color sensor.

Jumpers. 4 female-to-female for ultrasonic; 4 male-to-female for color.

For setup. A USB data cable, computer and the practice Hill. Optional: small cable ties and low-temperature hot glue.

8 jumper wires + 2 motor cables = 10 cables total.

WATCH THE BUILD VIDEO

Kit bot build video

Watch the full assembly once before you start, then follow the wiring pages.

BUILD-VIDEO LINK

youtube.com/watch?v=H7sx9fCWMdI

Scan the code or click the link.

Build in this order

1. Assemble the frame. Follow the video for the exact part placement. Seat the Technic pins and axles fully; wheels must turn freely.

2. Mount the electronics. Keep the Wukong stable and the micro:bit, power switch, USB ports and buttons accessible.

3. Wire with power off. Use the ultrasonic picture on page 2 and color-sensor picture on page 3.

4. Test, then secure. Run the programming guide's checks before adding glue. Finish with the tips on page 4.

Position the sensors

Ultrasonic faces forward. Keep both round metal faces clear of frame pieces and wires. Point it toward an approaching opponent, not down at the floor.

Color sensor faces down. Keep its sensing window and white LED unobstructed. It must see the black edge before the drive wheels reach the drop.

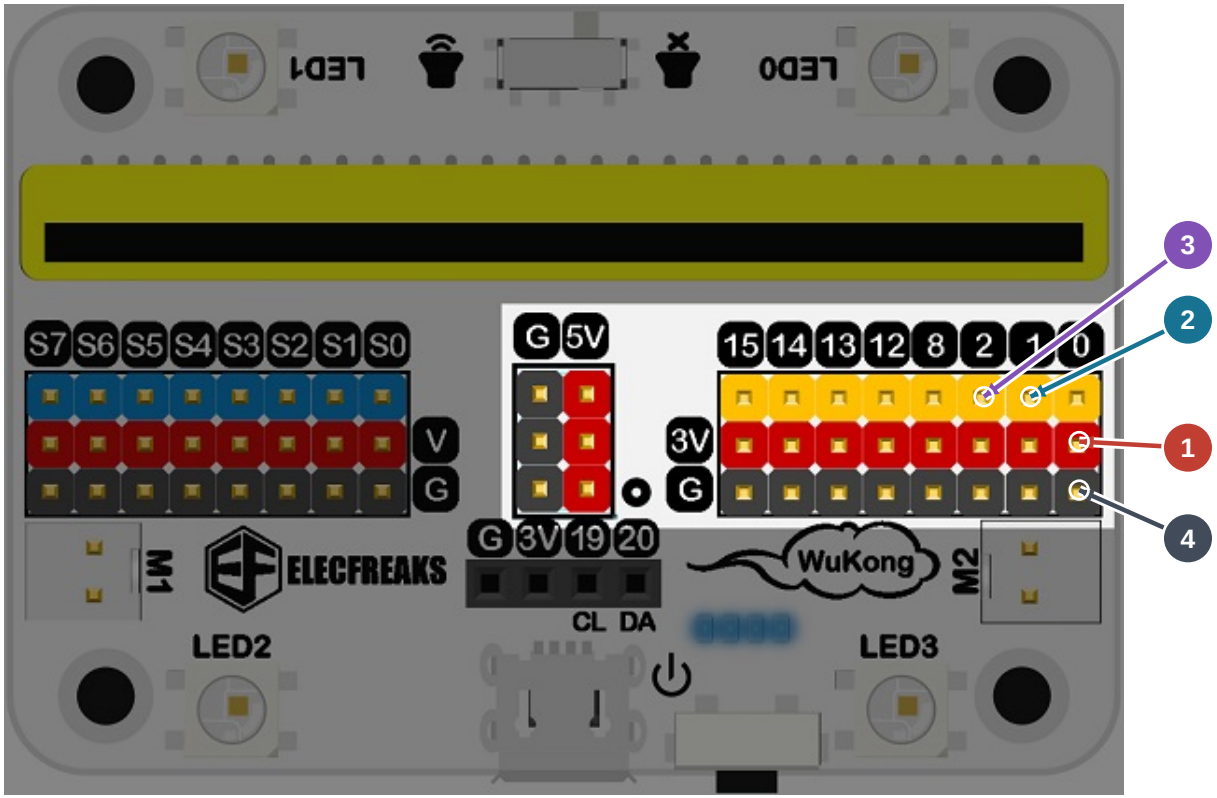
Keep the mounting height steady. Leave clearance for the field slopes. Recalibrate whenever the sensor height or angle changes.

Keep it serviceable. Leave enough slack to unplug a connector by its plastic housing. Do not pull on the wires. Take a clear wiring photo before adding glue.

Companion document: KitBot Programming Guide - project links, download steps, calibration and lessons.

1. Connect the ultrasonic sensor

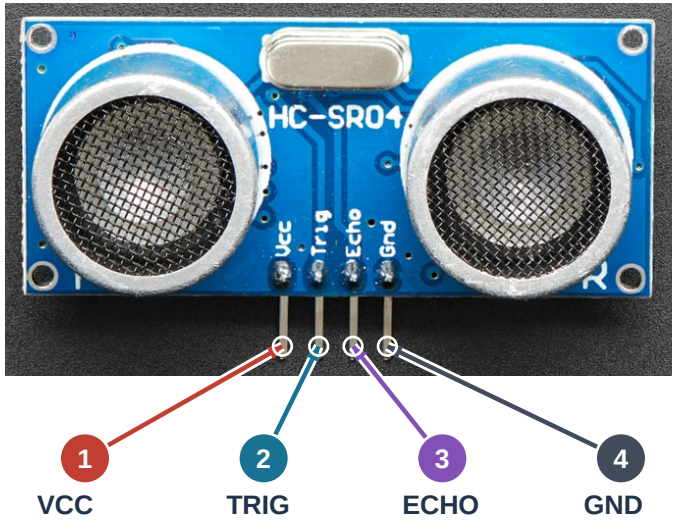
Use 4 female-to-female jumpers: a socket at both ends. Switch power off first.



Orient the Wukong with its micro:bit slot at the top, as pictured.

HC-SR04 ultrasonic sensor

Front view; read the labels above the pins.



Match each wire number to the table below.

Photo is a reference; follow your pin labels.

WIRE	SENSOR LABEL	EXACT WUKONG CONNECTION
1	**VCC**	Column 0 - RED pin in the 3V row
2	**TRIG**	Column 1 - YELLOW signal pin (P1 in MakeCode)
3	**ECHO**	Column 2 - YELLOW signal pin (P2 in MakeCode)
4	**GND**	Column 0 - BLACK pin in the G row

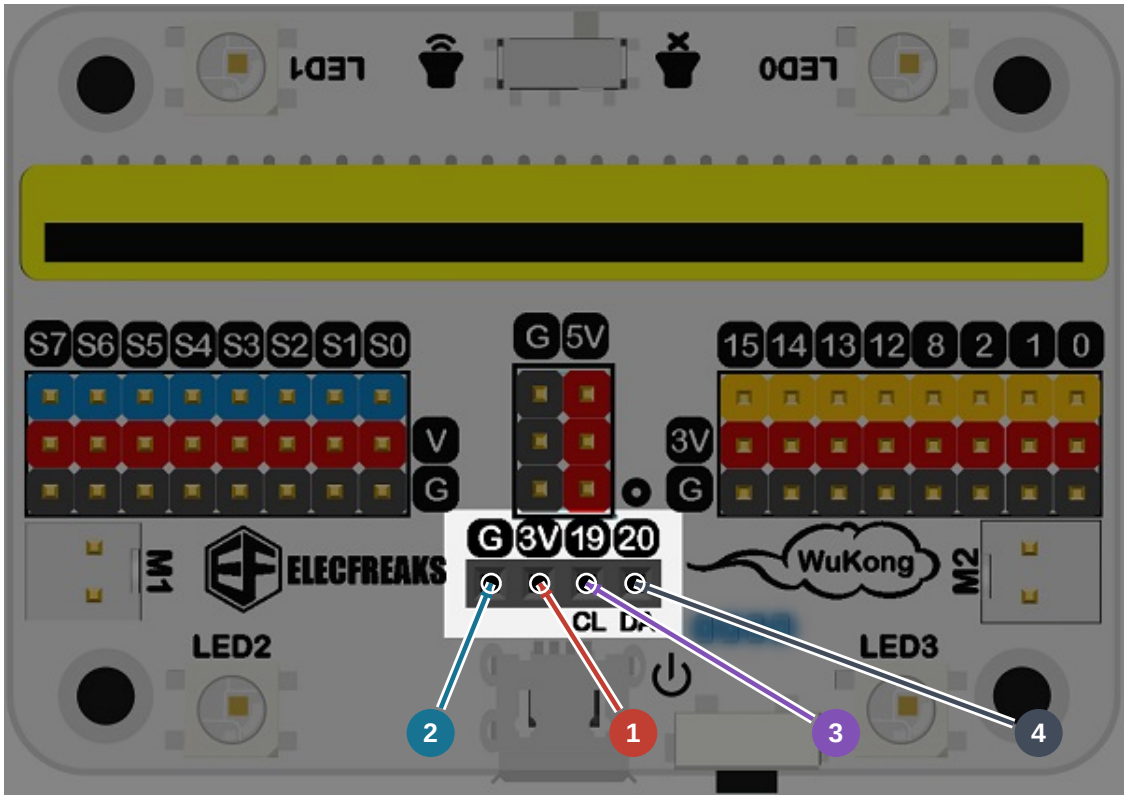
Leave yellow 0 empty. Do not use the blue S0/S1/S2 pins or the separate 5V pins in the middle.

This diagram uses 3V. Do not move VCC to 5V while ECHO is connected directly to pin 2.

Numbers identify the four wires. Your jumper insulation colors may differ from the arrow colors.

2. Connect the color sensor

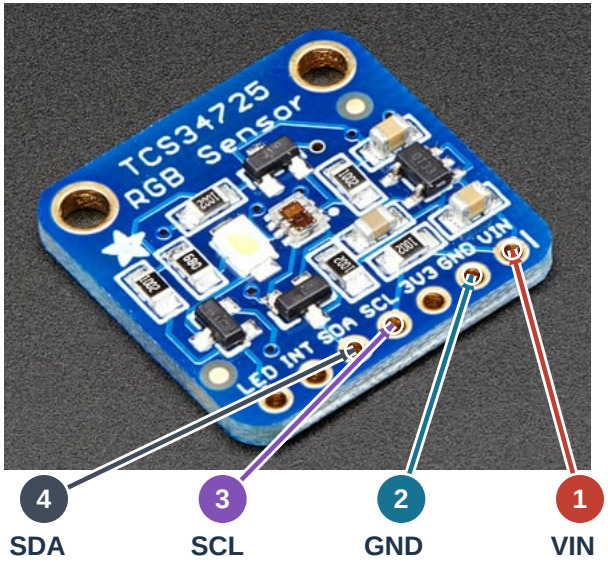
Use 4 male-to-female jumpers: metal-pin ends at the Wukong, socket ends at the sensor.



Use the four-hole socket labeled G 3V 19 20, just above the USB port.

TCS34725 color sensor

Use VIN, GND, SCL and SDA.



Reference photo shows the board before pins are fitted. Follow labels if your board differs.

WIRE	SENSOR LABEL	EXACT WUKONG CONNECTION
1	**VIN**	3V socket
2	**GND**	G socket
3	**SCL**	19 socket
4	**SDA**	20 socket

Leave LED, INT and the color sensor's 3V3 pin unconnected. Use VIN for the power wire.

Motors: one cable to M1, one to M2. Total: 8 sensor jumpers + 2 motor cables.

19 and 20 carry the color sensor's data connection, called I2C. Jumper insulation colors may vary.

Secure it. Check it. Then drive.

Use small, removable fixes so the robot survives handling and remains easy to repair.

Secure connections after testing

Seat each plug first. Check the pin label and gently press the connector fully onto the pin. Replace a loose or damaged jumper; glue will not repair its electrical contact.

Add a small retaining dab if needed. With power off, use low-temperature hot glue on the outside of a fully seated plastic connector housing and a nearby fixed support. Keep glue out of the socket and off the metal contacts.

Take strain off the pins. Secure the insulated wire to the frame nearby with a cable tie or a small dab of glue. Leave a short slack loop so a tug does not pull on the sensor header.

Let it cool before handling. Ask an adult for help with hot glue. Keep the hot nozzle and glue away from the battery, USB sockets, switches and buttons.

Mount sensors and route wires

Keep the ultrasonic steady. Use the frame or a bracket first. If it still moves, a small dab of low-temperature hot glue between a component-free board edge and the frame can help. Keep both round sensing faces uncovered.

Keep the color sensor adjustable. Test its height on every field ring before fixing it in place. Do not cover the sensing window, illumination LED or underside with glue.

Keep wires inside the frame. Route them clear of tires, axles, gears and the underside of the robot. Turn the wheels by hand to check for rubbing or snagging.

Keep tires clean. No glue on tire treads or axles. Trim loose glue strings and cable-tie ends; check that no wire or part drags on the field.

COMMON MISTAKE: REVERSED MOTORS OR SWAPPED M1/M2 PORTS

1 / Trace the motor cables

Switch power off. Follow each motor cable to M1 or M2 and compare with the build video. If the ports are swapped, correct them. Label the cables before securing them.

2 / Check motor direction

Driving backward or spinning during a forward command can mean reversed motor direction or swapped ports. Check mounting and port assignment first. Never flip a three-wire motor plug to reverse direction.

3 / Retest before driving

Raise both wheels; use Edge Starter over a non-black surface. Press A and wait five seconds. Both wheels should drive forward. Press B to stop. Power off before rewiring; correct one thing, reset and retest.

Before a match: secure parts, visible team number, red/blue marker, size/weight check, and working start/stop.

References: SRC Game Manual | Adafruit Wires and Connections